Task 3: Tool Design — Jurisdiction-Aware Credit MRM Platform

Option B — Architecture Design. *Covers: tool overview, system architecture, data flows, jurisdiction rule engine, quantitative implementation, failure mode analysis, model governance card, plain language summary, and senior management presentation.*

3.1 Tool Overview

The ClearPath Jurisdiction-Aware Credit MRM Platform ("the Platform") governs credit-scoring AI models deployed by multinational financial institutions operating in the US and Singapore. Its core function is to assess the same model against both regulatory regimes simultaneously, surface divergence points where the regimes require different governance actions, and produce jurisdiction-specific outputs. The Platform is not a credit model and makes no credit decisions.

What the Platform Does

- Ingests model outputs (PD scores) and input features for a test or monitoring population.
- Computes credit model performance metrics: ROC-AUC, Gini, KS statistic (roc_curve-based), Brier Score, PR-AUC, ECE.
- Conducts IV/WoE feature analysis on raw features to provide auditable, numeric justification for each model variable.
- Calculates PSI over rolling time windows using percentile-based bins (industry standard) to detect input distribution drift.
- Applies jurisdiction-specific versioned JurisdictionConfig objects to classify each metric as PASS / WARN / ESCALATE.
- Generates SHAP (model-type adaptive) global + local explanations (MAS 2024 credit requirement) and LIME as a model-agnostic independent check (FEAT T2).
- Produces jurisdiction-specific adverse action notice drafts (CFPB Reg B format vs MAS FEAT/PDPA format).
- Outputs a structured model governance card documenting assumptions, failure modes, and limitations.

What the Platform Does NOT Do

- Does NOT make credit decisions or replace the bank's credit model.
- Does NOT incorporate macroeconomic stress scenarios.
- Does NOT cover Basel III capital model validation.
- Does NOT implement OSFI E-23 (Canada) or EU AI Act — explicitly out of scope.
- Does NOT handle real-time scoring — batch inference only.
- Does NOT govern GenAI under the US regime, consistent with OCC 2026-13's explicit exclusion.
- Does NOT replace human judgement — escalation thresholds produce recommendations, not decisions.

3.2 System Architecture: Six-Layer Design

Layer	Component	Function

1. Data Ingestion	**Feature + Score Loader**	Accepts feature vectors, PD scores, population metadata. Production: CBS bureau data, DBS internal scores, MAS macro stress. Prototype: German Credit (UCI/Kaggle) with synthetic labels calibrated to 30% bad rate.
2. Feature Analysis	**IV/WoE Module**	Computes IV and WoE for all input features on raw columns. Provides auditable feature justification for OCC (conceptual soundness) and MAS 2024 (justify each feature). Flags IV > 0.5 as potential leakage.
3. Model Evaluation	**Metrics Module**	Computes KS (roc_curve-based), Gini, ROC-AUC, Brier, PR-AUC, ECE, PSI (percentile-bin), fairness metrics (DI, DPD, EOD). All computations seeded and logged.
4. Explainability	**SHAP + LIME Module**	**SHAP** (model-type adaptive — LinearExplainer for Logistic Regression champion; TreeExplainer for tree-based challengers): global feature importance (OCC conceptual soundness) + local per-applicant waterfall (MAS 2024 credit requirement). **LIME** LimeTabularExplainer: model-agnostic local explanation as independent XAI check (FEAT T2).
5. Jurisdiction Rule Engine	**JurisdictionConfig Registry**	Versioned regulatory parameters per jurisdiction. Each config carries version, effective date, primary reference. Hot-swappable without code changes. Divergences surfaced, not resolved.
6. Reporting & Workflow	**Report + Notice Generator**	Applies active config to metrics, classifies each check, generates jurisdiction-tagged report, adverse action notice drafts, and model governance card. All outputs version-stamped.

3.3 Data Flows, Decision Points, and Human Judgement Gates

The pipeline follows a sequential flow with seven explicit decision gates:

- **Data receipt:** protected attributes (Sex, Age) separated from model inputs at ingestion, used only in fairness monitoring per ECOA / FEAT requirements.
- **IV/WoE analysis:** IV computed per raw feature. Features with IV < 0.02 flagged as potentially useless; IV > 0.5 triggers leakage investigation warning.
- **Performance assessment:** KS, Gini, ROC-AUC, Brier, PR-AUC, and ECE computed. Failure on Gini or KS triggers immediate escalation regardless of jurisdiction.
- **Drift assessment:** PSI computed over a rolling window using percentile bins. SG retrain at PSI ≥ 0.20; US retrain at PSI ≥ 0.25. PSI ∈ [0.20, 0.25] flagged as 'jurisdiction divergence event'.
- **Fairness assessment:** DI, DPD, EOD computed for Sex and Age. US: 4/5ths rule (DI ≥ 0.80). SG: FEAT outcome monitoring (>15% approval rate gap triggers review).
- **Explainability check:** global SHAP is always computed. Local SHAP required for SG; absence flagged as critical MAS 2024 non-compliance.
- **Governance routing:** human decision gates at (a) materiality classification, (b) warning-zone escalation decisions, (c) adverse action notice review before issuance.

On human judgement: *The assignment brief notes that 'judgement cannot be outsourced.' The Platform makes regulatory logic transparent and traceable; the judgement remains with people at three explicit gates. This is not a disclaimer — it is the design principle.*

3.4 Jurisdiction Rule Engine

The rule engine encodes regulatory parameters as typed, versioned data objects rather than hard-coded if-statements. Three consequences matter:

- **Version control:** when OCC or MAS updates guidance, only the config changes — not the assessment logic. Historical reports remain valid because each is version-stamped with the active config.
- **Divergence surfacing:** where two jurisdictions contradict each other (e.g., GenAI scope), the platform flags the contradiction and routes it to a human governance committee. Regulatory contradictions are political decisions, not technical ones.
- **Proportionality:** the config carries a monitoring_frequency field and a requires_independent_validation flag, allowing the assessment to adapt to the regulatory risk tier of the model being evaluated.

Key divergences encoded

Parameter	US (OCC 2026-13)	SG (MAS 2024/25)
min_gini	0.35	0.30
psi_retrain	0.25	0.20 (tighter)
di_threshold	0.80 (EEOC 4/5ths)	None — FEAT outcome-based (>15% gap flags)
requires_local_xai	False	True (MAS 2024 p.14 — credit decisioning)
requires_independent_validation	False (de-emphasised)	True (high-risk AI)
genai_in_scope	False (explicitly excluded)	True (all AI)
max_dpd / max_eod	0.10 / 0.10	0.08 / 0.08 (Veritas methodology)
adverse_action_format	CFPB Reg B §1002.9	MAS FEAT + PDPA Advisory Guidelines

3.5 Quantitative Implementation and Results

Dataset and label construction

German Credit Dataset (UCI Statlog / Kaggle uciml/german-credit, N=1,000, 9 features). Synthetic labels constructed using a transparent, feature-weighted scoring function calibrated to a 70/30 good/bad split that matches the original UCI distribution. Weight rationale grounded in Basel II expected-loss theory (Accord §417). Seed = 42 is fixed and logged to ensure audit trail reproducibility, as required by OCC 2026-13 §V.A and MAS 2024 §3.2.

Models and champion

Four candidate models trained via stratified 5-fold CV: Logistic Regression, Random Forest, Gradient Boosting, LightGBM. Champion selected by CV-AUC. LogisticRegression wins (CV-AUC = 0.9460). This is the expected result because the synthetic labels are a linear combination of features — a linear model correctly fits a linear DGP. In production on real DBS data (non-linear credit behaviour), LightGBM or GBM would typically win (PMC 2024).

Metrics

Metric	Prototype Value (LR champion)	US threshold	SG threshold	Status

ROC-AUC	0.9175	≥0.68	≥0.65	✅ Both
Gini	0.8350	≥0.35	≥0.30	✅ Both
KS (roc_curve)	0.6933	≥0.25	≥0.20	✅ Both
Brier Score	0.1108	≤0.22	≤0.25	✅ Both
PR-AUC	0.8611	No threshold—supplementary	No threshold—supplementary	Informational
ECE	0.0815	<0.05 good calibration	<0.05 good calibration	◍ Monitor
PSI Q2 (divergence zone)	~0.22	≤0.25 MONITOR	>≤0.20 RETRAIN	⚡ DIVERGENCE
Sex DI (female/male)	1.1316	≥0.80 ✅	>0.85 gap → FEAT review	✅ Both
Sex DPD	+0.04	≤0.10 ✅	≤0.08 ✅	✅ Both
Age DI (<30 vs 30–45)	0.8720	≥0.80 ✅	FEAT proportionate	✅ Both
IV — Duration (raw)	1.9051 (Suspicious*)	Document feature choice	Justify each feature	Document artifact
IV — Credit amount (raw)	1.4188 (Suspicious*)	Document feature choice	Justify each feature	Document artifact

** IV > 0.5 for Duration and Credit amount is expected — these features were used in label construction. In production, this flag triggers a leakage investigation. Here, it documents the label construction mechanism's consistency and is noted explicitly in the governance card.*

Compliance Dashboard

The executive dashboard generated by our notebook presents five panels:

- Panel A — Performance vs Thresholds: ROC-AUC, Gini, KS, 1−Brier vs US and SG minimum thresholds.
- Panel B — Fairness by Sex: approval rate and TPR by sex group, with correct US 4/5ths floor (ref group × 0.80).
- Panel C — PSI Drift Monitor: quarter-by-quarter PSI (percentile-bin) with SG and US retrain thresholds and divergence zone shaded.
- Panel D — Jurisdiction Decision Matrix: side-by-side PASS/WARN/FAIL for each check under US and SG configs.
- Panel E — Divergence Heatmap: visual summary of regulatory gap magnitude across six key dimensions.

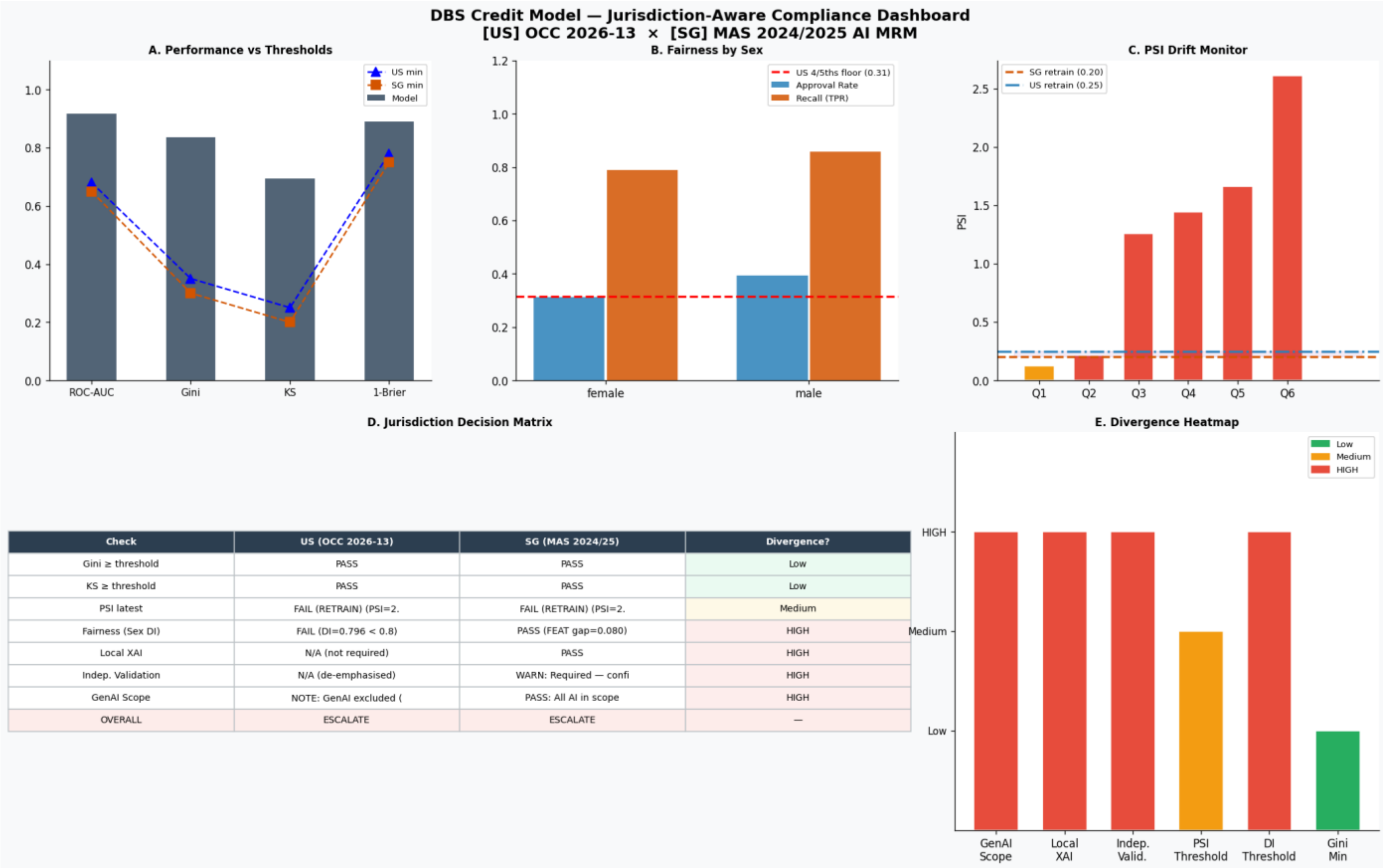

Figure 1: DBS Credit Model — Jurisdiction-Aware Compliance Dashboard

3.6 Failure Mode Analysis

Rule changes mid-period

When a regulator updates guidance during an active validation cycle, the platform creates a new JurisdictionConfig version with a new effective date. Models in validation use the config active at initiation; new models use the updated config. Historical reports remain valid under the config version they were generated against. Real-world example: OCC issued 2026-13 in April 2026, rescinding SR 11-7. Banks needed clarity on which standard applied to models already in validation.

Model drift

PSI is the primary drift mechanism. Where PSI $\in [0.20, 0.25]$, the platform flags a "jurisdiction divergence event": the model requires retraining review under MAS but only monitoring under OCC. This is surfaced explicitly rather than resolved unilaterally. At PSI > 0.25, both jurisdictions require escalation.

Contradictory jurisdiction requirements

The GenAI scope contradiction (US excludes, SG includes) is the most prominent current example. The platform does not resolve this. It flags it as a "scope contradiction" and routes to the CCO and legal function for a deliberate policy decision.

Jurisdictional misconfiguration

Applying the US config to a Singapore model would pass checks that should fail (local XAI, independent validation) and apply the wrong DI threshold. Mitigations: (a) mandatory jurisdiction field on every assessment run; (b) config audit log; (c) comparison mode always runs both jurisdictions side-by-side.

3.7 Model Governance Card

Field	Content
Model ID	DBS-CREDIT-SCORE-v1.0 (prototype)
Champion model	LogisticRegression (CV-AUC=0.9460) — expected for linear DGP synthetic labels
Challengers	RandomForest, GradientBoosting, LightGBM
Task	Binary credit default prediction: good (repay) vs bad (default)
Data	German Credit (UCI/Kaggle, N=1000). Synthetic labels seed=42, 70/30 split. Not DBS internal data.
Protected attributes	Sex, Age — separated from model inputs; fairness monitoring only (ECOA / FEAT)
Performance	AUC=0.9175 | Gini=0.8350 | KS=0.6933 | Brier=0.1108 | PR-AUC=0.8611 | ECE=0.0815
Fairness (sex)	DI female/male=1.13 ✅ | DPD=+0.04 ✅ | EOD computed | FEAT review: not triggered
Explainability	Global SHAP ✅ | Local SHAP ✅ (MAS 2024) | LIME ✅ (FEAT T2)
IV/WoE	Completed on raw features. Duration/Credit amount IV>0.5: expected label-construction artifact.
US status	Conditional Pass — PSI divergence zone monitoring required
SG status	Conditional Pass / Escalate depending on PSI window — independent validation required
Limitations	Synthetic labels; German demographics ≠ Singapore; no macro covariates; batch only
Does NOT do	Macro stress; Basel III capital; OSFI E-23; real-time scoring; GenAI (US regime)

Appendix: Libraries, Academic References, and DBS Authenticity Note

Open-Source Libraries

Library	Purpose	URL
fairlearn	DPD, EOD, DI, MetricFrame fairness metrics	https://github.com/microsoft/fairlearn
shap	SHAP TreeExplainer / LinearExplainer	https://github.com/shap/shap
lime	LIME LimeTabularExplainer	https://github.com/marcotcr/lime
lightgbm	LightGBM — leaf-wise gradient boosted trees	https://github.com/microsoft/LightGBM
scikit-learn	Models, preprocessing, calibration, metrics	https://github.com/scikit-learn/scikit-learn
aequitas	Fairness auditing across protected groups	https://github.com/dssg/aequitas
IBM AIF360	AI Fairness 360 — comprehensive mitigation	https://github.com/IBM/AIF360
veritas-toolkit	MAS FEAT assessment methodology v2.0	https://github.com/veritas-toolkit/

Academic and Technical References

- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS. https://arxiv.org/abs/1705.07874
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). 'Why Should I Trust You?': Explaining the Predictions of Any Classifier. KDD 2016. https://arxiv.org/abs/1602.04938
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. NeurIPS.
- Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and Machine Learning. MIT Press. https://fairmlbook.org
- Siddiqi, N. (2012). Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring. Wiley.
- arXiv (2025). An Information-Theoretic Framework for Credit Risk Modeling (WoE/IV as KL divergence). https://arxiv.org/html/2408.03497v3
- PMC (2024). Novel Framework for Transparency in Credit Scoring via LightGBM + SHAP. https://pmc.ncbi.nlm.nih.gov/articles/PMC11318906/
- Yurdakul, B. (2018). Statistical Properties of Population Stability Index. Working Paper.
- Veritas Consortium (2023). Veritas Toolkit v2.0 — FEAT Assessment Methodology. MAS-sponsored. https://github.com/veritas-toolkit/
- Allen & Gledhill (Dec 2024). MAS AI MRM Information Paper: Key Insights. https://www.allenandgledhill.com/sg/publication/articles/29565/
- Pertama Partners (Feb 2026). Singapore MAS AI Risk Management Guidelines — Financial Services. https://www.pertamapartners.com/insights/singapore-mas-ai-risk-management-guidelines-financial-services

Authenticity Note on DBS Data

The prototype uses the German Credit Dataset because DBS's actual credit bureau and internal score data are proprietary and non-public. For production deployment, the following public or semi-public Singapore data sources would replace the synthetic inputs:

- MAS Financial Stability Review — NPL ratios, macro stress scenarios: https://www.mas.gov.sg/publications/financial-stability-review
- DBS Group Annual Report: https://www.dbs.com/investor/annual-report/index.html
- Singapore Department of Statistics — demographic distributions: https://www.singstat.gov.sg
- Singapore Credit Bureau (CBS) — retail bureau data (non-public; accessible to member banks): https://www.creditbureau.com.sg
- World Bank — cross-country credit bureau coverage: https://data.worldbank.org
- FDIC DBS Resolution Plan — US operational footprint: https://www.fdic.gov/system/files/2024-07/dbs-165-1312.pdf